

# Shreejit Verma Homework 4

Thursday, 19 August 2021

9:02 AM

Q31

The proportion of defective products coming out of a production line,  $P$ , is a random variable, and varies day to day according to a Beta distribution with parameters  $\alpha = 1$  and  $\beta = 9$ . In other words, each day has its own value of  $P$ , drawn from the specified distribution. On a randomly chosen day 100 products are sampled and inspected. Let  $X$  be the number of defective products out of 100. Find  $E(X)$  and  $V(X)$ .

$$E(P) = \frac{\alpha}{\alpha + \beta} = \frac{1}{1+9} = \frac{1}{10}$$

$$V(P) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{1 \times 9}{(1+9)^2(1+9+1)}$$

$$= \frac{9}{(10)^2(11)} = \frac{9}{1100} \text{ Ans}$$

100 products are randomly chosen ( $X$  out of)

$$E\left(\frac{X}{100}\right) = E(P)$$

$$\Rightarrow \frac{1}{100} E(X) = E(P) \Rightarrow E(X) = 100 \times E(P)$$

$$= 100 \times \frac{1}{10} = 10$$

$$V\left(\frac{X}{100}\right) = V(P)$$

$$\Rightarrow \frac{V(X)}{(100)^2} = V(P) \Rightarrow V(X) = \frac{9}{1100} \times 100 \times 100$$

$$= \frac{900}{11} \text{ Ans}$$

33. Consider the following situation:

- $Y \sim N(\mu, \sigma^2)$
- $(X | Y = y) \sim N(y, \tau^2)$  [FYI:  $\tau$  is the Greek letter Tau]

**We want to find the distribution of  $X$ .**

One strategy would be to find  $f_{XY}(x, y)$  and then use a Law of Total Probability

$$f_X(x) = \int_y \underbrace{f_{X|Y}(x | y) f_Y(y)}_{f_{XY}(x, y)} dy$$

While this is fine, it requires some algebraic manipulations that may not be obvious.

Another strategy is to find the MGF  $M_X(t)$  and use it to identify the distribution of  $X$ :

$$M_X(t) = E(e^{tX}) = E[E(e^{tX} | Y)]$$

where the "inner" expected value,  $E(e^{tX} | Y)$  is the moment generating function of the random variable  $X | Y = y$ .

**Find  $M_X(t)$  and use to determine the distribution of  $X$ .**

$$M_X(t) = E(e^{tX}) = E[E(e^{tX} | Y)]$$

$$E(E(e^{tX} | Y)) = E(e^{Yt + \tau^2 \times \frac{t^2}{2}})$$

$$= e^{\tau^2 \frac{t^2}{2}} E(e^{Yt})$$

$$= e^{\tau^2 \frac{t^2}{2}} e^{\mu t + \sigma^2 \frac{t^2}{2}}$$

$$= e^{t^2 \frac{\tau^2 + \sigma^2}{2} + \mu t} \text{ (Ans)}$$

$$= \text{MGF of } N(\mu, \sigma^2 + \tau^2)$$

Q35

Assume that  $X$ ,  $Y$ , and  $Z$  are all random variables. Use our measure-theoretic notion of conditional expectation to explain why

$$E(Y | X) = E(E(Y | X, Z) | X).$$

$$E(E(Y | X, Z)) = E(Y)$$

$$E(E(Y | X, Z) | X) \left\{ \because X \subseteq X, Z \right\}$$

$$= E(Y | X) \left\{ \because E(E(X | Z) | Z) = E(X | Z) \right\}$$

42. Let  $\mathbf{X}$  be multivariate normal:  $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ , with

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0 & 0.5 \\ 0 & 1 & 0 \\ 0.5 & 0 & 1 \end{bmatrix}.$$

- What is the joint distribution of  $(X_2, X_3)$ ? Are  $X_2$  and  $X_3$  independent?
- Find  $E(X_2^2 \cdot e^{X_3})$
- Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

What is the distribution of  $\mathbf{Y} = \mathbf{A}\mathbf{X}$ ? Are the components of  $\mathbf{Y}$  independent of each other?

Soln<sup>n</sup> a)  $(X_2, X_3)$  is bivariate normal with  $\mu_Y = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

$$\text{and } \Sigma_Y = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

We see that  $\text{Cov}(X_2, X_3) = 0$  so  $X_2, X_3$  are independent since they are bivariate normal

$$\textcircled{b) } E(X_2^2 \cdot e^{X_3}) = E(X_2^2) \cdot E(e^{X_3}) \xrightarrow{M_{X_3}(t=1)}$$

$$= (\text{Var}(X_2) + E(X_2)^2) \cdot E(e^{X_3})$$

$$= (1 + (2)^2) \cdot (e^{\mu t + \sigma^2 \frac{t^2}{2}})$$

$$= (5) \cdot e^{(0 \times t + 1 \times \frac{1}{2})}$$

$$= 5 \cdot e^{\frac{1}{2}} \text{ Ans}$$

$$\textcircled{c) } Y = AX$$

$$E(Y) = E(AX) = A E(X) = A(\boldsymbol{\mu})$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 1 + 0 \times 2 + 1 \times 0 \\ 1 \times 1 + 0 \times 2 + 2 \times 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$V(Y) = V(AX) = A V(X) A^T = A \Sigma A^T$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0.5 \\ 0 & 1 & 0 \\ 0.5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1.1 + 0.0 + 1.0 & 1.0 + 0.1 + 1.0 & 1.0 \times 0.5 + 0.0 + 1.1 \\ 1.1 + 0.0 + 2.0 & 1.0 + 0.1 + 2.0 & 1.0 \times 0.5 + 0.0 + 2.1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1.5 & 0 & 1.5 \\ 2 & 0 & 2.5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1.5 \times 1 + 0.0 + 1.5 \times 1 & 1.5 \times 1 + 0.0 + 1.5 \times 2 \\ 2 \times 1 + 0.0 + 2.5 \times 1 & 2 \times 1 + 0.0 + 2.5 \times 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 4.5 \\ 4.5 & 7 \end{bmatrix}$$

No, components of  $Y$  are not independent of each other.